

Nihar Masurkar

+1 602 475 4920 | nmasurka@asu.edu | <https://www.linkedin.com/in/NiharMasurkar>

Summary

Ph.D. candidate with extensive expertise in bio-inspired robotics, control systems, robotic manipulation, as well as ultrasonic and acoustic testing.

Education

Ph.D. in Robotics and Automation Engineering

2023 - Present

Specialization: Non-Destructive Testing

GPA: 3.75

Arizona State University, Mesa, Arizona

Master of Science in Engineering, Robotics & Autonomous Systems

2019 - 2022

Concentration: Mechanical & Aerospace

GPA: 3.46

Arizona State University, Tempe, Arizona

Academic Projects & Research

Robotic Integration for Automated NDE, ISNDE Lab, ASU

Jan 2023 – Present

- Incorporating ultrasonic testing and air-coupled transducers into diverse robotic manipulators and mobile platforms, utilizing fiducial markers to mirror a digital twin of a pipeline.
- Establishing a motion capture system capable of precisely validating the joint trajectory of these robotic platforms to enhance and optimize the control system.

Bio-Inspired Tap-Scanning Robot, ISNDE Lab, ASU

Jan 2023 – Present

- Conducted motion analysis on the third digit of an Aye-aye (*Daubentonia madagascariensis*) finger to elucidate its dynamic range, distinguishing it as a distinctive biological tap scanner.
- Conceptualized a two-link robotic arm emulating the motion trajectory of the Aye-aye, with enhancements to the control system enabling force-based control of the robotic arm to perform tap scanning of cavities within wooden structures.

Octopus-Inspired Robotic Gripper, Academic Project: Bio-Inspired Robotics, ASU

Jan 2021 – May 2021

- Engineered a soft pneumatic modular gripper design inspired by the functional aspects of elephant trunks and octopus tentacles, prioritizing efficient object grasping with minimal contact area.
- Programmed a pneumatic controller for the silicone arms with central air channels. Successful testing revealed that a two-finger 200mm gripper could hold 45N in a maximum pull-off test and achieve whole 360-degree curling.

Lizard Inspired Tube Inspection Robot, BIRTH Lab, ASU

Jan 2020 – Dec 2023

- Developed a PID controller for a three-axis Cartesian robotic platform equipped with a load cell to assess the frictional characteristics of the gripper pad, essential for the functionality of the lizard-inspired tube inspection robot. This robot demonstrates the capability to grip the outer surface of the pipe and enable linear motion along its length.
- Manufactured customized frictional pads utilizing Polydimethylsiloxane (PDMS) embedded in the grippers, with optimization being a critical aspect given the friction-based mobility of the robot. A closed-loop controller was also designed to operate the gripper fingers, allowing for precise control of opening, closing, speed, load torque, and position using encoder feedback.

Publications

- Nemati, H., **Masurkar, N.**, & Dehghan-Niri, E. (2025). Investigating Acoustic-based Foraging Behavior in Aye-ayes (*Daubentonia madagascariensis*) Through Infrared Thermography. *International Journal of Primatology*, 46(3), 763–784. <https://doi.org/10.1007/s10764-025-00498-y>
- Nasr, M., **Masurkar, N.**, Meshkizadeh, P., & Dehghan-Niri, E. (2025). Guided by Hearing: Robotic motion planning for precision in gas leakage localization. *Materials Evaluation*, 83(4), 51–61. <https://doi.org/10.32548/2025.me-04491>
- **Masurkar, N.**, Das, A., Rudraboina, M., Morris-Sjolund, D., Alvidrez, F., Dehghan-Niri, E., & Marvi, H. (2024). A biologically-inspired tube inspection robot with friction-based mobility. *International Journal of Intelligent Robotics and Applications*, 9(1), 164–175. <https://doi.org/10.1007/s41315-024-00370-x>.
- **Masurkar, N.**, Jiming, K., Nemati, H., & Dehghan-Niri, E. (2023). Aye-aye middle finger kinematic modeling and motion tracking during tap-scanning. *Biomimetic Intelligence and Robotics*, 3(4), 100134. <https://doi.org/10.1016/j.birob.2023.100134>
- Hespeler, S., Nemati, H., **Masurkar, N.**, Alvidrez, F., Marvi, H., & Dehghan-Niri, E. (2023). Deep Learning–Based Time-Series classification for robotic inspection of pipe condition using Non-Contact Ultrasonic testing. *Journal of Nondestructive Evaluation, Diagnostics and Prognostics of Engineering Systems*, 7(1). <https://doi.org/10.1115/1.4063694>
- Nemati, H., Alvidrez, F., Das, A., **Masurkar, N.**, Rudraboina, M., Marvi, H., & Dehghan-Niri, E. (2021). Integrating electromagnetic acoustic transducers in a modular robotic gripper for inspecting tubular components. *Materials Evaluation*, 79(7), 715–727. <https://doi.org/10.32548/2021.me-04223>

Conferences

- E. Dehghan-Niri, **N. Masurkar**, H. Nemati, Z. Goode, H. S. Halliday and J. Liebig, "Overcoming Challenges in the Biomimetic Study of Termite Drumming Behavior," *2024 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, Bangkok, Thailand, 2024, pp. 172-177, <https://doi.org/10.1109/ROBIO64047.2024.10907719>.
- **N. Masurkar**, H. Nemati, and E. Dehghan-Niri "Enhancing biomimetic design of tap scanning sensors through high-resolution thermal camera-based behavioral studies", *Proc. SPIE 12944, Bioinspiration, Biomimetics, and Bioreplication XIV*, 129440E (9 May 2024); <https://doi.org/10.1117/12.3027268>
- **Masurkar, N.**, Nemati, H., Dehghan-Niri, E., "Advancing the Biomimetic Design for Automated Tap Scanning Through Aye Aye's Behavioral Studies Utilizing High-Resolution Thermal Cameras", *Proceedings of the 2024 51st Annual Review of Progress in Quantitative Nondestructive Evaluation*. 2024 51st Annual Review of Progress in Quantitative Nondestructive Evaluation. Denver, Colorado, USA. July 22–24, 2024.
- **Masurkar, N.**, Meshki Zadeh, P., Dehghan-Niri, E., "Automated Robotic Approach for Source Localization of Minor Gas Leak Using a Two-Microphone Array", *32nd ASNT Research Symposium*, Pittsburgh, PA, June 24th - 28th, 2024.
- **Masurkar, N.**, Nemati, H., Dehghan-Niri, E., "Enhancing the biomimetic design of tap scanning sensors through high-resolution thermal camera-based behavioral studies", *SPIE, Smart Structures/NDE*, 300 Long Beach Blvd, Long Beach, CA, 2024.
- **Masurkar, N.**, Nemati, H., Dehghan-Niri, E., & Hespeler, S., "An automated robotic approach for inspection of metallic tubular structure", *31st ASNT Research Symposium*, Columbus, Ohio, June 26th – 30th, 2023. <https://doi.org/10.32548/rs.2023.003>
- Nemati, H., Alvidrez, F., Das, A., **Masurkar, N.**, Rudraboina, M., Marvi, H. & Dehghan-Niri, N., "Toward Automated Ultrasonic Inspection of Pipelines and Tubular Components", *ASNT Research Symposium*, Virtual, March 2021

Technical Skills

- **Languages:** Java, C++, C, Python
- **Electronic Design:** ExpressPCB, Autodesk Eagle CAD, KiCAD, Arduino IDE, MATLAB
- **Design and Application:** Autodesk Fusion 360, Solidworks
- **Other Skills:** Amazon EC2, Amazon IoT Core, Amazon Kinesis Stream
- **Certifications:**
 - Control of Mobile Robots by Prof. Magnus Egerstedt
 - PCB Design and Fabrication using EagleCAD

Coursera, June 2019
NPTEL, January 2017

Professional Experience

IoT Developer, University Technology Office, ASU

Aug 2020 – May 2023

- Devised an algorithm to cluster objects within a point cloud captured by Ouster OS0-32 LIDAR, then processed these clusters through a machine learning model for classification and detection. This facilitated the assessment of population density around security poles on campus and the monitoring of vehicular traffic in their vicinity.
- Engineered an electrical circuit integrated with diverse environmental sensors to survey the surroundings of a security pole. This circuit efficiently transmits data streams to AWS IoT Core via MQTT protocol for rapid and efficient data transmission.
- Configured Power over Ethernet (PoE) IP cameras to stream live video footage from security poles onto a webpage using Amazon Video Kinesis. Additionally, I developed an optimized Multiple Object Tracking (MOT) script employing a YOLO model for detecting and classifying multiple individuals around security poles.

Awards

- Achieved 2nd place in Devils Invent: Protecting America's Public Access Areas competition as part of [Team BAHMMN](#). Developed an early response system utilizing dynamic exit signage systems to disperse crowds efficiently.
- Earned 1st place in the Starbucks Hackathon for designing an immersive Augmented Reality Experience tailored for café baristas to view order details via an Augmented Reality display.
- Clinched 1st place in Devils Invent: Drone Defense, a hackathon sponsored by the Department of Homeland Security and Boeing, organized by ASU. Designed a system to safeguard significant buildings from potential aerial threats as a member of [Team KCAaN](#).
- Secured funding for the Lizard-Inspired Tube Inspection Robot project through ASU Engineering's Research Program (MORE) and garnered special sponsorship from W.L Gore & Associates.